

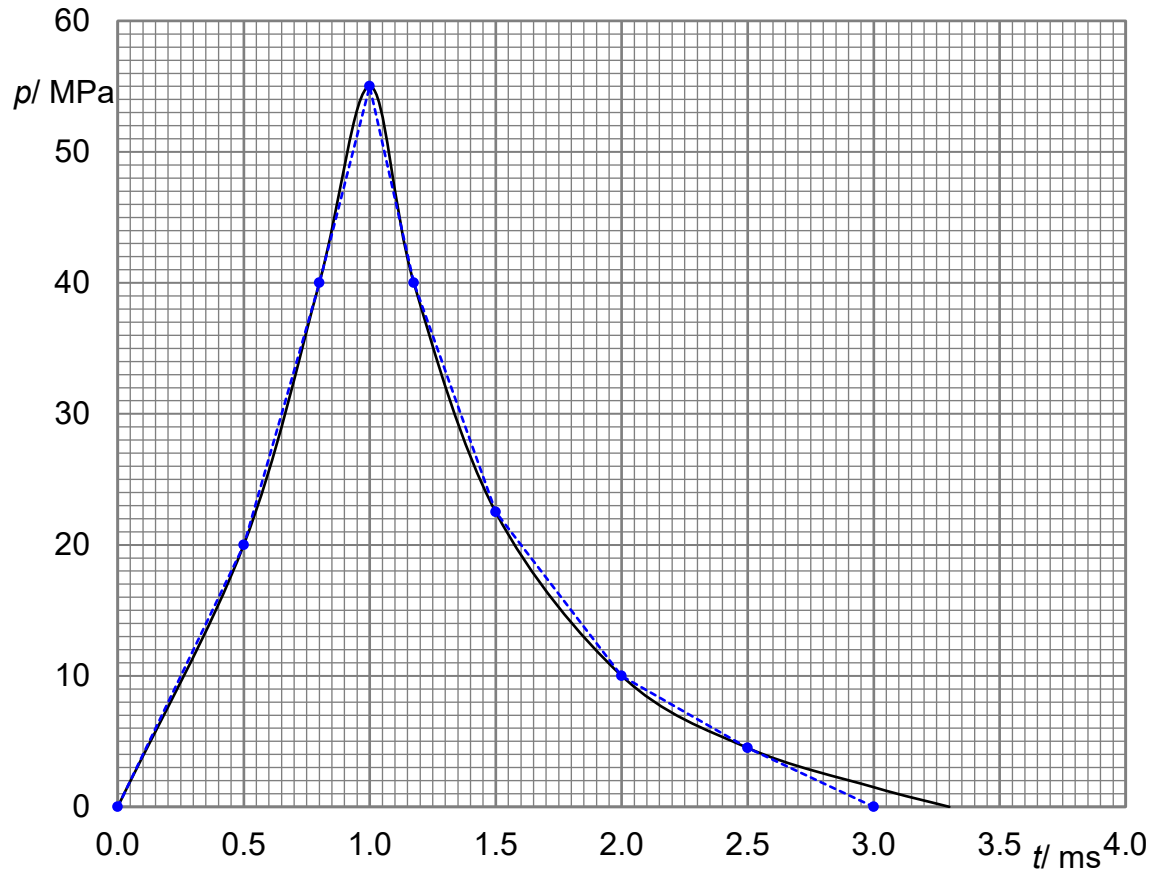
Tutorial 6 Collisions Suggested Solutions

D1 (a) (i) From the graph, $p_{\max} = 55.0 \times 10^6 \text{ Pa}$

$$F_{\max} = p_{\max} A = 55.0 \times 10^6 (2.8 \times 10^{-4}) = 1.54 \times 10^4 \text{ N}$$

(ii)
$$a_{\max} = \frac{F_{\max}}{m} = \frac{1.54 \times 10^4}{3.2 \times 10^{-2}} = 4.8 \times 10^5 \text{ m s}^{-2}$$

(b)



Using trapezium rule,

Area under $p - t$ graph

$$\begin{aligned} &= \frac{1}{2} [20.0 + (20.0 + 55.0) + (55.0 + 22.5) + (22.5 + 10.0) + (10.0 + 4.5) + 4.5] \times 10^6 \times 0.500 \times 10^{-3} \\ &= (20.0 + 55.0 + 22.5 + 10.0 + 4.5) \times 10^6 \times 0.500 \times 10^{-3} = 5.6 \times 10^4 \text{ Pa s} \end{aligned}$$

Impulse on the projectile

$$I = \text{Area under the } p - t \text{ graph} \times A_{\text{cross-sectional}} = 5.6 \times 10^4 \times 2.8 \times 10^{-4} = 16 \text{ N s}$$

OR

Counting squares,

$$\text{Area under } p - t \text{ graph} = 11 \times (10.0 \times 10^6) \times 0.500 \times 10^{-3} = 5.5 \times 10^4 \text{ Pa s}$$

Impulse on the projectile

$$I = \text{Area under the } p - t \text{ graph} \times A_{\text{cross-sectional}} = 5.5 \times 10^4 \times 2.8 \times 10^{-4} = 15 \text{ N s}$$

(c) Change in momentum

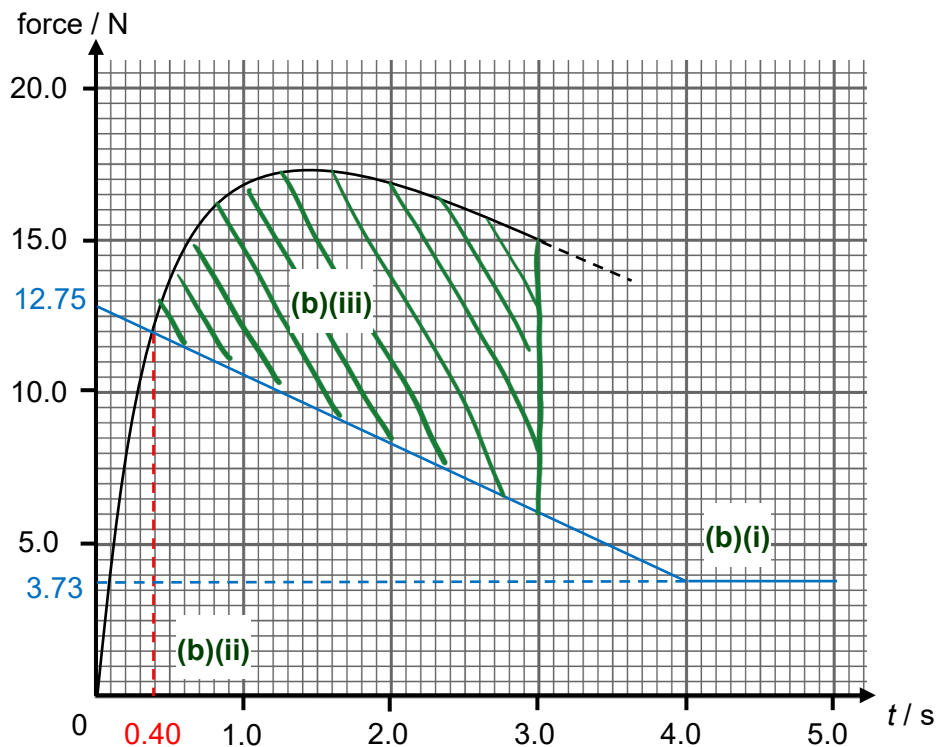
$$\Delta p = m(v_f - v_i) = 3.2 \times 10^{-2}(270 - 0) = 8.6 \text{ N s}$$

(d) The answer in (c) is smaller than the answer in (b). This is due to friction between the barrel and the projectile. The resultant force acting on the projectile is smaller, leading to a smaller change in momentum of the projectile.

D2 (a) Time taken for fuel to be burn

$$t = \frac{\Delta m}{\Delta m / \Delta t} = \frac{1.30 - 0.38}{0.23} = 4.0 \text{ s}$$

(b) (i)



Initial weight of the rocket

$$W_{\text{initial}} = 1.30 \times 9.81 = 12.75 \text{ N}$$

Final weight of the rocket

$$W_{\text{final}} = 0.38 \times 9.81 = 3.73 \text{ N}$$

(horizontal and vertical axes values may be read to 2 d.p.)

For $0 \leq t \leq 4.0$, straight line graph with a negative gradient.

For $4.0 \leq t \leq 5.0$, horizontal line since no change in weight.

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- (ii) Rocket only lifts off when upward force exceeds weight of the rocket. i.e., at point of intersection of both lines.

From the graph, $t = 0.40 \text{ s}$

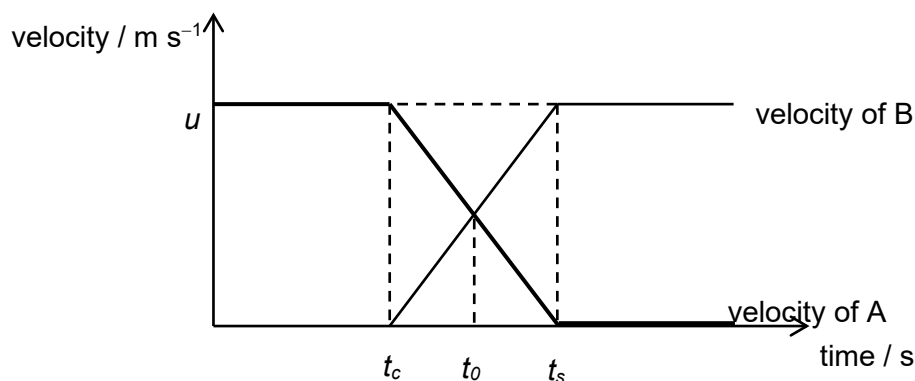
- (iii) From Newton's Second Law,

$$F_R = \frac{dp}{dt} \Rightarrow \Delta p = \int F_R dt \text{ where } F_R = F - W$$

Shade area between both lines from the lift-off ($t = 0.4 \text{ s}$) to $t = 3.0 \text{ s}$

- D3** (a) Body A will be at rest and body B will move with velocity u in the direction of the initial velocity of body A.

- (b)



Allow straight line as time of collision is very small.

- (c) If body A has a higher velocity than body B, there will be further compression, i.e. not maximum compression yet. If body A has a lower velocity than body B, then body B will move away from body A, i.e. they will separate.
- (d) By conservation of linear momentum,

$$mu + 0 = (2m)v_0$$

$$v_0 = \frac{1}{2}u$$

- (e) At t_o ,

$$E_{0,kinetic} = \frac{1}{2}(2m)\left(\frac{1}{2}u\right)^2 = \frac{1}{4}mu^2$$

After they have completely separated,

$$E_{s,kinetic} = \frac{1}{2}(m_B)(u_B)^2 = \frac{1}{2}mu^2$$

- (f) Kinetic energy at maximum compression is less than kinetic energy after separation. Some kinetic energy is transformed into the elastic potential energies stored in the compression of the two bodies at maximum compression.

- D4 (a)** By principle of conservation of momentum,

$$mu + 0 = 2mv$$

$$v = \frac{1}{2}u$$

- (b)** Initial KE of system

$$KE_{before} = \frac{1}{2}mu^2$$

Final KE of system

$$KE_{after} = \frac{1}{2}(2m)\left(\frac{1}{2}u\right)^2 = \frac{1}{4}mu^2$$

Ratio

$$\frac{KE_{after}}{KE_{before}} = \frac{\frac{1}{4}mu^2}{\frac{1}{2}mu^2} = 0.5$$

- (c)** The 'lost' kinetic energy is converted into other forms such as thermal energy and sound energy.

- D5 (a)** When the spring is at maximum compression, the 2 blocks move together with the same velocity v , beyond which the blocks start to separate and move at different speed.

By principle of conservation of momentum,

$$p_{initial} = p_{final, x_{max}}$$

$$(2.0)(10) + (3.0)(5.0) = (2.0 + 3.0)v_{final}$$

$$v_{final} = 7.0 \text{ m s}^{-1}$$

By conservation of energy,

$$E_{initial} = E_{final, x_{max}}$$

$$\frac{1}{2}(2.0)(10)^2 + \frac{1}{2}(3.0)(5.0)^2 = \frac{1}{2}(7.0)^2(2.0 + 3.0) + \frac{1}{2}(1000)x^2$$

$$x = 0.173 \text{ m}$$

- (b)** When the spring re-extends to its original length, all the elastic PE stored in it is transferred back to the blocks as KE, as if it is an elastic collision.

Assuming the blocks both move off to the right after the collision, by the principle of conservation of momentum,

$$(2.0)(10) + (3.0)(5.0) = 2.0v_A + 3.0v_B$$

$$35 = 2.0v_A + 3.0v_B \text{ ----- (1)}$$

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Since collision is perfectly elastic,
relative speed of approach = relative speed of separation

$$10.0 - 5.0 = v_B - v_A$$

$$v_B = v_A + 5.0 \text{ ----- (2)}$$

Solving, subst (2) into (1)

$$35 = 2.0v_A + 3.0(v_A + 5.0)$$

$$v_A = 4.0 \text{ ms}^{-1} \text{ (to the right)}$$

$$v_B = 9.0 \text{ ms}^{-1} \text{ (to the right)}$$

C1 By Conservation of Momentum,

Horizontally:

$$m_1 v_1 = m_1 v_1' \cos \theta_1 + m_2 v_2' \cos \theta_2 \text{ ----- (1)}$$

Vertically:

$$m_1 v_1' \sin \theta_1 = m_2 v_2' \sin \theta_2$$

$$v_2' = \frac{m_1 v_1' \sin \theta_1}{m_2 \sin \theta_2} \text{ ----- (2)}$$

Subst (2) into (1):

$$m_1 v_1 = m_1 v_1' \cos \theta_1 + m_2 \cos \theta_2 \left(\frac{m_1 v_1' \sin \theta_1}{m_2 \sin \theta_2} \right)$$

$$v_1 = v_1' \cos \theta_1 + v_1' \cot \theta_2 \sin \theta_1$$

$$\cot \theta_2 = \frac{v_1 - v_1' \cos \theta_1}{v_1' \sin \theta_1}$$

$$\tan \theta_2 = \frac{v_1' \sin \theta_1}{v_1 - v_1' \cos \theta_1} \quad (\text{shown})$$

Since $v_1 = v_1' \cos \theta_1 + v_1' \cot \theta_2 \sin \theta_1$

$$\frac{v_1'}{v_1} = \frac{1}{\cos \theta_1 + \cot \theta_2 \sin \theta_1}$$

% loss in KE

$$= \frac{E_{k,initial} - E_{k,final}}{E_{k,initial}}$$

$$= \frac{\frac{1}{2} m_1 v_1^2 - \frac{1}{2} m_1 v_1'^2}{\frac{1}{2} m_1 v_1^2}$$

$$= 1 - \frac{v_1'^2}{v_1^2}$$

$$= 1 - \left(\frac{1}{\cos \theta_1 + \cot \theta_2 \sin \theta_1} \right)^2$$

$$= 0.128$$

$$= 12.8\%$$